

CART-POLE WITH A LIGHT CART

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Motivation

A cart sliding freely on a horizontal track carries a pendulum. When the cart is much lighter than the pendulum bob, the mass matrix of the system becomes nearly singular every time the pole passes through the vertical: its determinant falls by six orders of magnitude, twice per swing. The near-singularity comes from the geometry, not from a modelling choice. A nearly massless cart cannot resist the horizontal reaction of the pole, so at the vertical almost nothing opposes horizontal motion.

The consequence is a spike in the velocities. Each time the pole passes the vertical, $\dot{\theta}$ rises from a few rad/s to about 4,429 rad/s and falls back again within roughly 1.08 μ s, while the swing itself has a period of about 1.8s. The two time scales differ by more than a factor of a million. The system is a two-body tree with no closed loops, no constraints and no contact, so all of the difficulty comes from the conditioning of the mass matrix and the speed of these events.

The problem is meant to test three things: whether step-size control locates and resolves the events; whether the method computes accelerations accurately when the mass matrix is near-singular; and whether a solution reported as successful is in fact accurate. On the last point, a standard explicit integrator run at a relative tolerance of 10^{-6} , already tighter than the default in most ODE libraries, completes the problem and reports success with no warning, yet its error is about a hundred times the acceptance threshold defined below. The problem complements the *Double four-bar mechanism*, in which singular positions arise in the constraints rather than in the mass matrix, and follows the *Stiff flyball governor* [1] in its definition of the reference solution and the error.

System

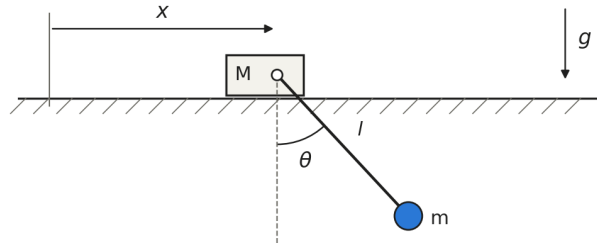


Figure 1: Cart-pole. The cart (mass M) slides without friction along the horizontal track. A massless rigid pole of length l , pinned to the cart, carries a point mass m at its tip.

The system moves in the vertical plane. The cart position x is measured along the track, and the pole angle θ is measured from the downward vertical, positive when the bob is on the $+x$ side of the cart. The generalised coordinates are $q = (x, \theta)$. The bob lies at $(x + l \sin \theta, -l \cos \theta)$, and gravity acts in the negative vertical direction. No other forces act: there is no friction, no damping and no actuation.

Cart mass	M	$1 \times 10^{-6} \text{ kg}$
Bob mass (point mass at the pole tip)	m	1 kg
Pole length (massless, rigid)	l	1 m
Gravity	g	9.81 m s^{-2}
Initial cart position and velocity	$x_0, \dot{x}_0$	$0 \text{ m}, 0 \text{ m s}^{-1}$
Initial pole angle and angular velocity	$\theta_0, \dot{\theta}_0$	$\pi/2 \text{ rad (horizontal)}, 0 \text{ rad s}^{-1}$
Total simulation time	T	10 s

For reference, the equations of motion are $\mathbf{M}(q) \ddot{q} = \mathbf{F}(q, \dot{q})$ with

$$\mathbf{M}(q) = \begin{bmatrix} M + m & ml \cos \theta \\ ml \cos \theta & ml^2 \end{bmatrix}, \quad \mathbf{F}(q, \dot{q}) = \begin{bmatrix} ml\dot{\theta}^2 \sin \theta \\ -mgl \sin \theta \end{bmatrix}, \quad \det \mathbf{M} = ml^2 (M + m \sin^2 \theta). \quad (1)$$

The determinant ranges from $ml^2(M + m)$ when the pole is horizontal down to ml^2M when it is vertical, a ratio of $(M + m)/M \approx 10^6$.

Because no horizontal force acts and the system starts at rest, the horizontal momentum $(M + m)\dot{x} + ml\dot{\theta} \cos \theta$ remains zero. The centre of mass therefore never moves horizontally, and the bob, which carries almost all of the mass, travels almost vertically. The bob stays at a horizontal position of about l while the cart slides beneath it between $x \approx 0$ and $x \approx 2l$. The pole passes through the vertical 11 times in 10 s, first at $t = 0.451525 \text{ s}$.

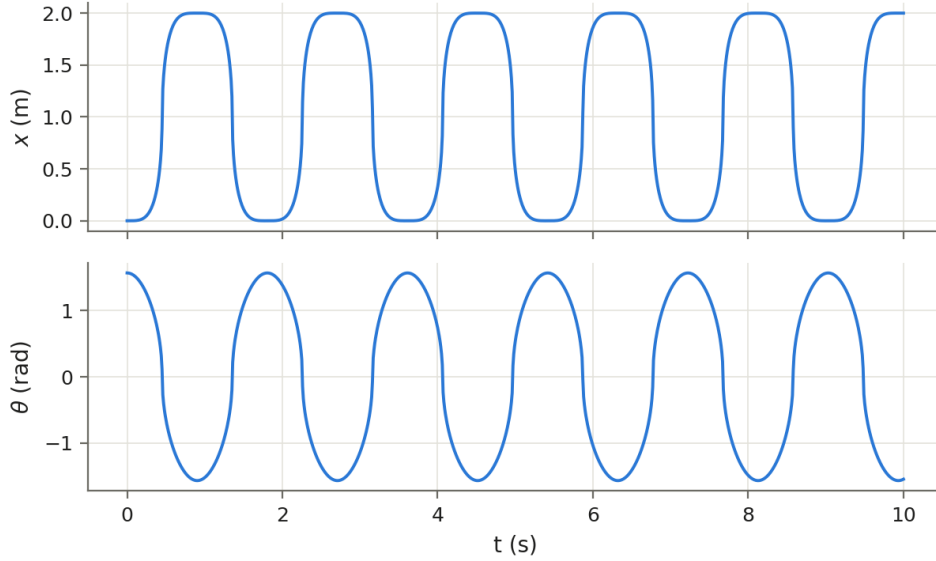


Figure 2: Reference time histories of the cart position x and the pole angle θ .

Reference solution

The reference solution was not produced by any multibody package. The equations of motion were derived by hand from the Lagrangian, and the 2×2 system was solved in closed form:

$$\ddot{x} = \frac{m \sin \theta (l\dot{\theta}^2 + g \cos \theta)}{M + m \sin^2 \theta}, \quad \ddot{\theta} = -\frac{\sin \theta ((M + m)g + ml\dot{\theta}^2 \cos \theta)}{l(M + m \sin^2 \theta)}. \quad (2)$$

The only division is by $M + m \sin^2 \theta$, a sum of non-negative terms, so no cancellation occurs however light the cart becomes. These equations were integrated with the explicit eighth-order Runge–Kutta method DOP853 [2] as implemented in SciPy [4]. The relative tolerance was tightened from 10^{-4}

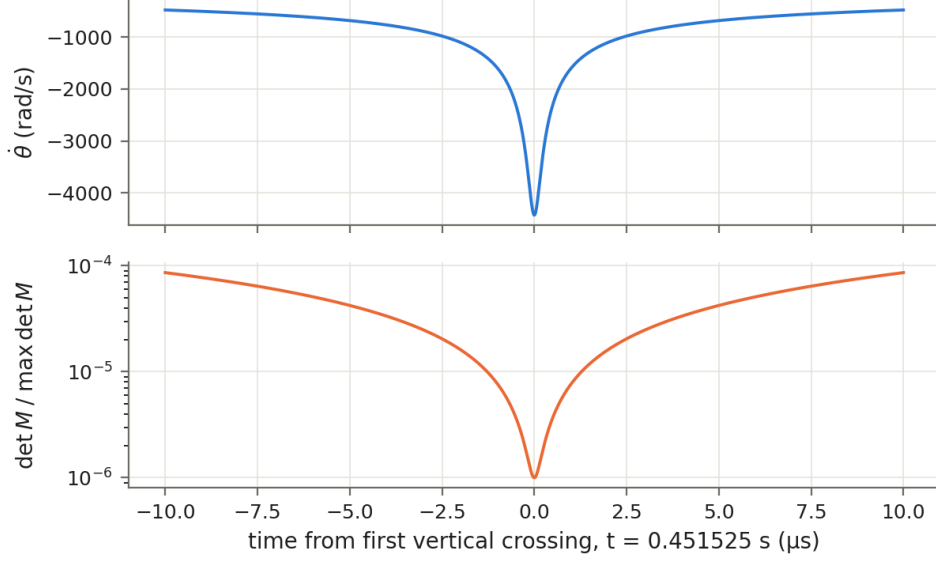


Figure 3: Angular velocity, and determinant of the mass matrix relative to its maximum, in the $20\ \mu\text{s}$ around the first vertical crossing. The spike has a peak of $4,429\ \text{rad/s}$ and a full width at half maximum of $1.08\ \mu\text{s}$; the same event recurs at every crossing.

to 10^{-13} (with absolute tolerance $10^{-2} \times$ the relative tolerance), and the run at 10^{-13} forms the reference. Repeating the ladder with the implicit Radau IIA method [3] gives the same converged trajectory.

The convergence was checked against a solution obtained without any time integration. With zero horizontal momentum and conserved energy, the motion reduces to the quadrature

$$\dot{\theta}^2 = \frac{2g(M+m)(\cos\theta - \cos\theta_0)}{l(M+m\sin^2\theta)}, \quad x = \frac{ml(\sin\theta_0 - \sin\theta)}{M+m}, \quad (3)$$

which was evaluated, and inverted for $\theta(t)$, in 30-digit arithmetic with mpmath [5]. It gives a quarter period of $0.4515245357\ \text{s}$. Against this solution the integrated reference has an error e (defined below) of 9.1×10^{-12} , and its 11 vertical crossings fall within $1.9 \times 10^{-12}\ \text{s}$ of the exact times.

Integrator	Accelerations	rtol	e vs. quadrature	$\max \Delta E $ (J)	Evaluations
DOP853	closed form	10^{-4}	7.3×10^{-3}	4.7×10^{-3}	10,352
DOP853	closed form	10^{-6}	1.1×10^{-4}	1.4×10^{-4}	15,923
DOP853	closed form	10^{-8}	1.5×10^{-7}	3.5×10^{-7}	25,574
DOP853	closed form	10^{-10}	2.0×10^{-9}	2.5×10^{-9}	43,154
DOP853	closed form	10^{-12}	1.2×10^{-10}	1.4×10^{-10}	53,240
DOP853	closed form	10^{-13}	9.1×10^{-12}	8.8×10^{-12}	70,214
Radau	closed form	10^{-8}	1.2×10^{-7}	1.4×10^{-7}	124,746
Radau	closed form	10^{-10}	7.2×10^{-10}	9.9×10^{-10}	386,046
Radau	closed form	10^{-13}	6.8×10^{-11}	2.0×10^{-11}	2,046,293
DOP853	linear solve	10^{-8}	1.5×10^{-7}	3.5×10^{-7}	25,562
DOP853	linear solve	10^{-13}	3.1×10^{-9}	2.6×10^{-9}	70,373

Table 1: Convergence of the integrated solution as the tolerance is tightened. e is the benchmark error measured against the 30-digit quadrature solution; $\max|\Delta E|$ is the largest departure of the mechanical energy from its initial value. “Linear solve” obtains the accelerations by numerically solving $\mathbf{M}\ddot{\mathbf{q}} = \mathbf{F}$ rather than from the closed form; it stops improving at about 10^{-9} , around three hundred times worse than the closed form.

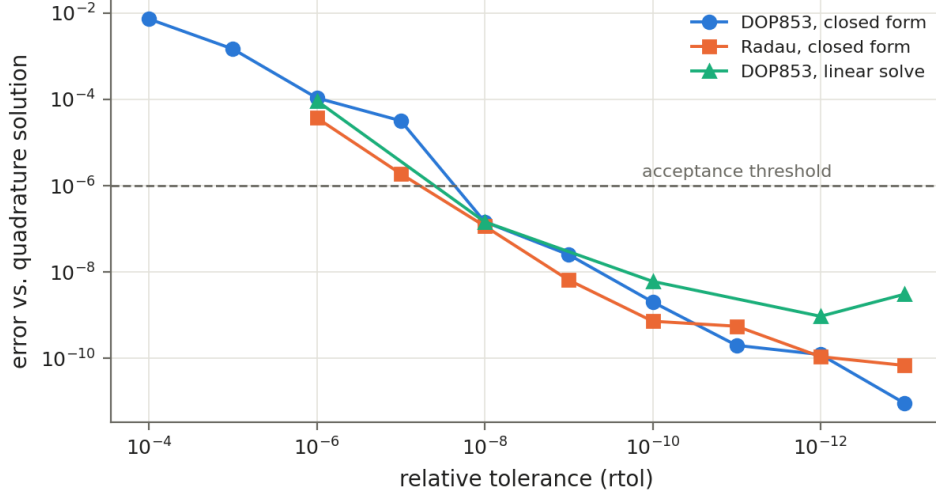


Figure 4: Error against relative tolerance for each integrator and formulation. The dashed line is the acceptance threshold.

The energy appears in the results file only as a secondary check. It converges together with the trajectory (Table 1), but a small energy drift does not by itself show that a solution is accurate. The determining check is convergence of the positions to the quadrature solution.

Error definition and objective

Solutions are compared with the reference at the $n = 1001$ sampling times $t_i = 0.01(i - 1)$ s, $i = 1, \dots, n$. The error of a simulation is

$$e = \sqrt{\frac{1}{n} \sum_{i=1}^n \left[\left(\frac{x(t_i) - x_{\text{ref}}(t_i)}{l} \right)^2 + (\theta(t_i) - \theta_{\text{ref}}(t_i))^2 \right]}. \quad (4)$$

Absolute rather than relative errors are used because θ passes through zero. Velocities are not scored. The 0.01 s grid never lands inside a spike, so velocity samples would mostly measure timing jitter; an error committed inside a spike shows up instead as a shift in the timing of x and θ , which e does capture.

A simulation is considered correct if $e \leq 10^{-6}$. The objective is to run the 10 s simulation in the minimum CPU time while keeping e below this threshold. Submitters are encouraged to also report the energy drift $\max_i |E(t_i) - E(0)|$ and, for methods that use them, the number of function evaluations.

Results file

The reference file `results.txt` has a single header line followed by 1001 rows, one per sampling time, with six whitespace-separated columns:

1. time t (s), from 0 to 10 in steps of 0.01;
2. cart position x (m);
3. pole angle θ (rad), measured from the downward vertical and not wrapped;
4. cart velocity $\dot{x}$ (m/s);

5. pole angular velocity $\dot{\theta}$ (rad/s);
6. mechanical energy relative to its initial value, $E(t) - E(0)$ (J).

Submitted results must contain at least the first three columns at the same sampling times; the remaining columns are optional. The script that generates the reference solution, the quadrature check and every figure is provided in the accompanying archive and at <https://github.com/Noah-Parsons/light-cartpole-benchmark>.

References

- [1] M. González, D. Dopico, U. Lugić, and J. Cuadrado. A benchmarking system for MBS simulation software: problem standardization and performance measurement. *Multibody System Dynamics*, 16(2):179–190, 2006.
- [2] E. Hairer, S. P. Nørsett, and G. Wanner. *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, 2nd ed., 1993.
- [3] E. Hairer and G. Wanner. *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*. Springer, 2nd ed., 1996.
- [4] P. Virtanen et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.
- [5] The mpmath development team. mpmath: a Python library for arbitrary-precision floating-point arithmetic, version 1.3.0, 2023. <https://mpmath.org>.